

Memory: Feeding AI's Voracious Hunger for Data

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OBJECTIVE ANALYSIS



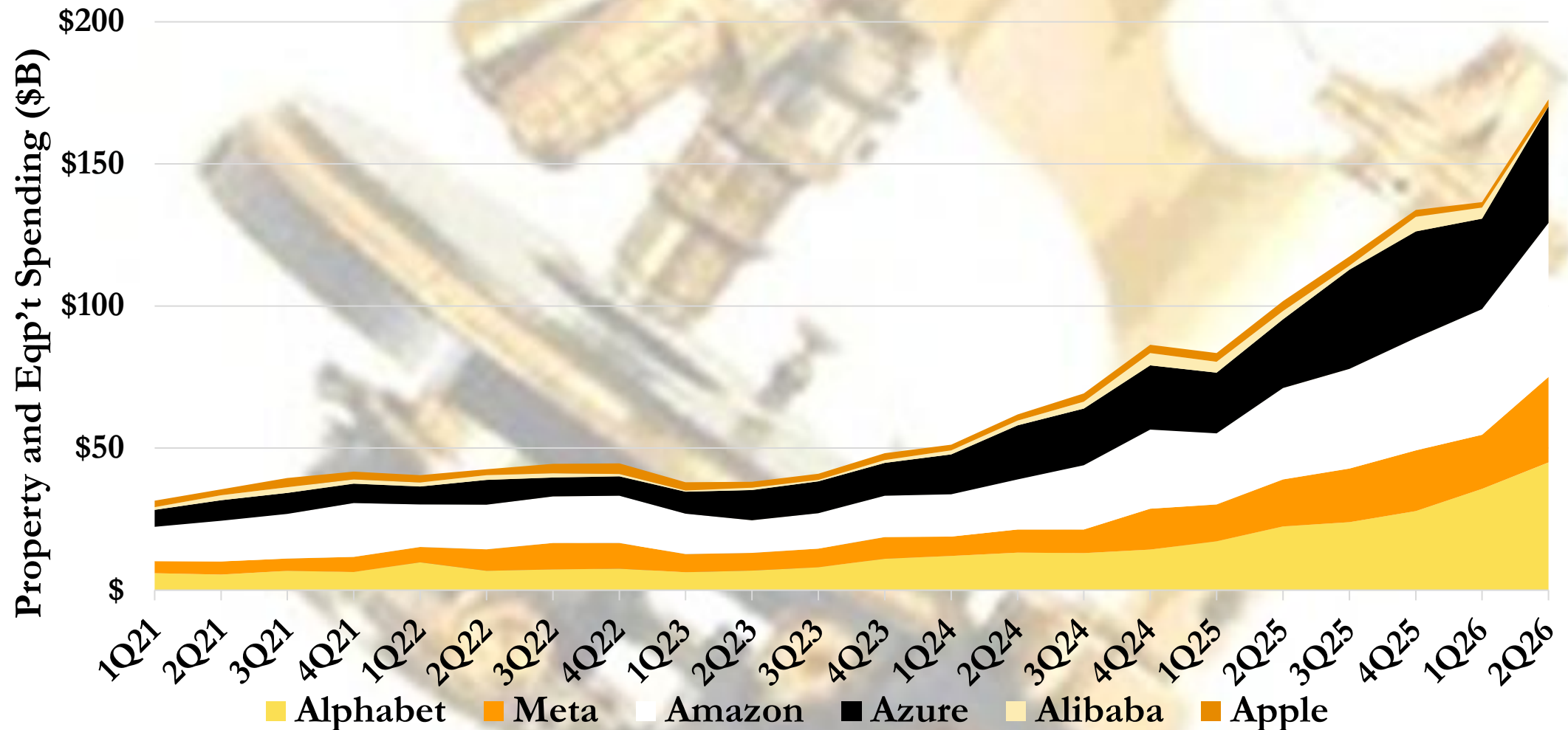


Agenda

1. The current AI memory market
2. Future directions & solutions
3. Today's instructors

Hyperscalers' Capital Spending Binge

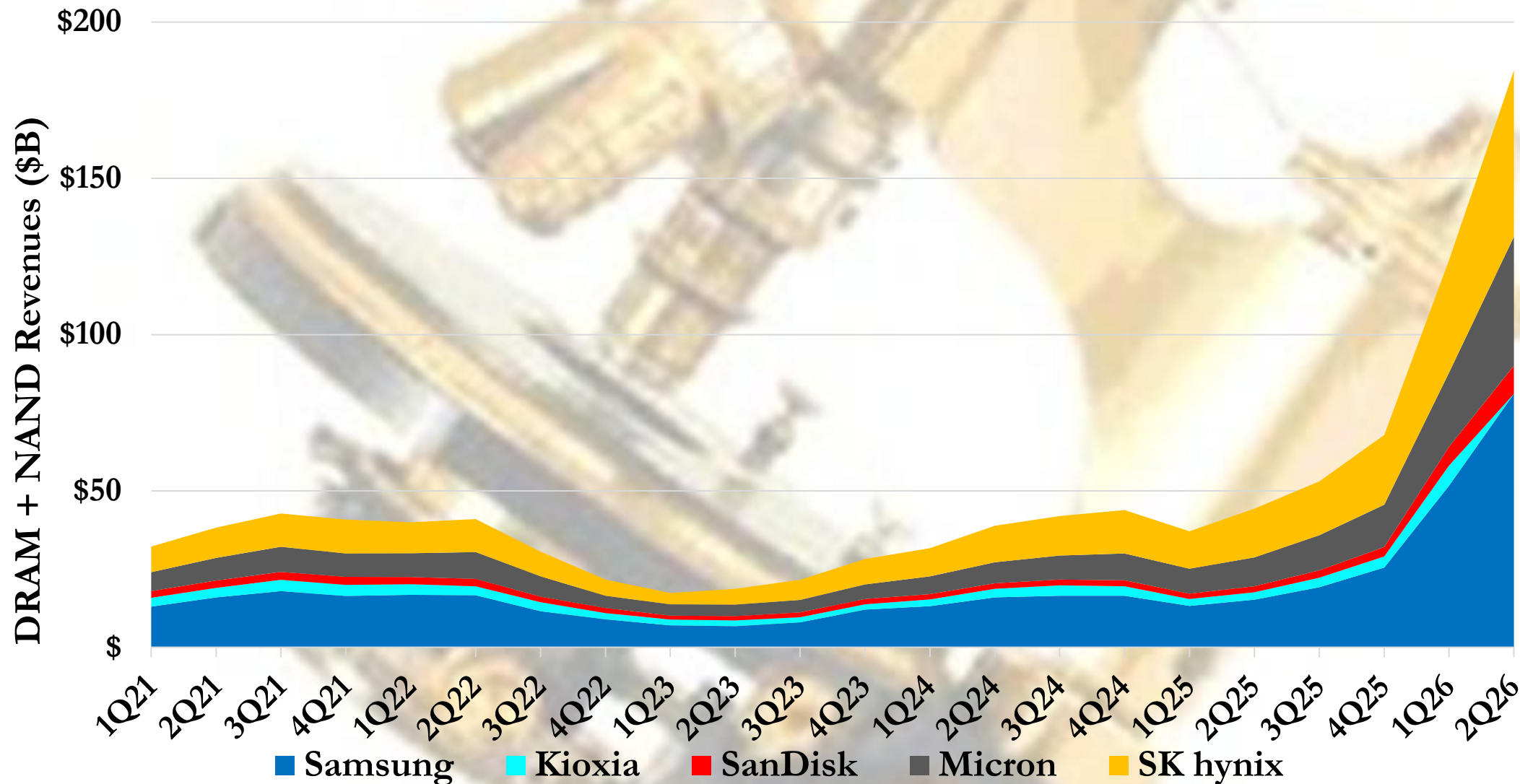
Hyperscale Datacenter Capital Spending



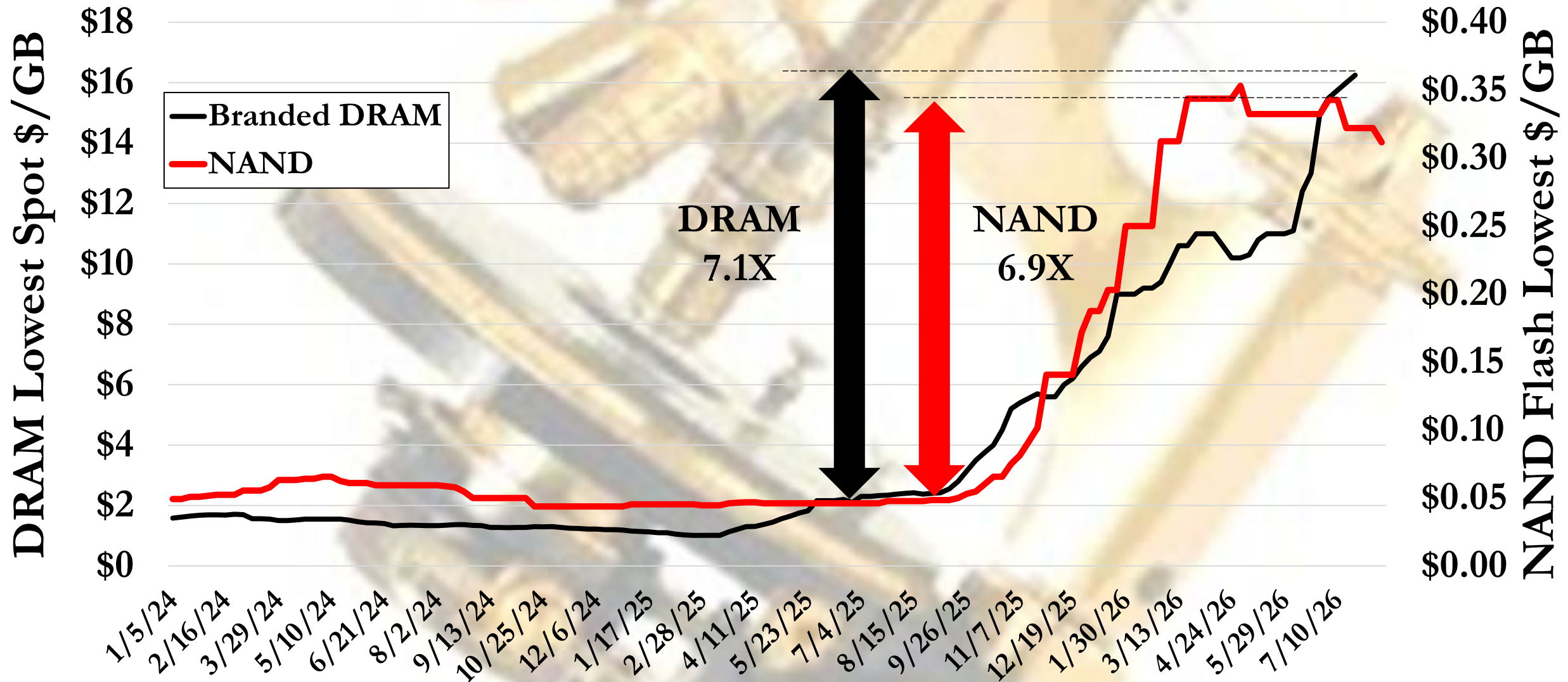
3 Drivers of DRAM Shortage

- AI spending more than anticipated
- Bandwidth requirements consume inordinate wafer volume
 - HBM takes 3X the die area of DDR
- DRAM wafer volume static for >10 years
 - New capacity takes >2 years to install

Result: Record Memory Revenues



Most of Growth is from Pricing



HBM & DRAM Future Dynamics



- New report from Objective Analysis
- Covers all perspectives
 - What is HBM and why it's so costly
 - Supply Chain – who buys it and why
 - Price dynamics: HBM vs. DRAM
 - Forecast (Revenues, units, ASP)
- Downloadable in September:
 - [Objective-Analysis.com/reports](https://www.Objective-Analysis.com/reports)



2. Long-Term Outlook

Moving to Inference

- Inference resources span a very wide range
 - Microcontrollers sufficient for the lowest end
 - Hyperscale datacenters remain at the top (LLMs)
- Ongoing algorithm refinements
 - Today's data movement issues may fade out
- Opportunities for processing in memory (PIM)

Processing In/Near Memory

- Use the base die in HBM
- Custom memory chips for higher performance
- Analog Neural Nets (ANN) for the low end
- Software support is currently weak
 - This is a gating factor

Could Ease Memory Wall Issues

2026 Memory Wall Looks Different



Reducing Data Traffic

The bandwidth requirement of a communications channel is inversely proportional to the intelligence at either end



“AI Everywhere”

- AI will become invisible
- AI will drive down costs
- Everyone will use agents all the time
- Similar to earlier technology adoptions:
 - Semiconductors
 - Microprocessors & microcontrollers
 - The Internet



What's Being Done About It?



3. Your Instructors

Raghu Sreeramaneni, Micron

Evolving memory architectures for AI



Fellow, HBM Design Architecture
Micron Technology

Sangwook Han, PhD, Samsung

HBM Base Die: How will HBM evolve in the future by utilizing the logic process?



Design Lead, Custom HBM4E
Samsung Electronics

Jaesik Lee, SK hynix

Advanced Packaging for High Bandwidth Memory (HBM)



VP of Package Engineering
SK hynix America

D-Matrix: 3D DRAM based Accelerator for Generative Inference

Sudeep Bhoja



Founder/CTO, D-Matrix

Aayush Ankit



Software Engineer, Meta

Oxmiq Labs: HBF in AI Compute: A System Architect's view

Anurag Agrawal



System Architect, Oxmiq

Radhakrishna Giduthuri



Architect, Praxmati